

Luiss
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Numerical System

Introduction to Computer Programming

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Numerical System

A Numerical System is a writing system used to represent numeric values with symbols called digits in a consistent manner.

The number of digits used to express number is called base or radix.

Decimal number system has **base 10** and uses digits from 0 to 9.

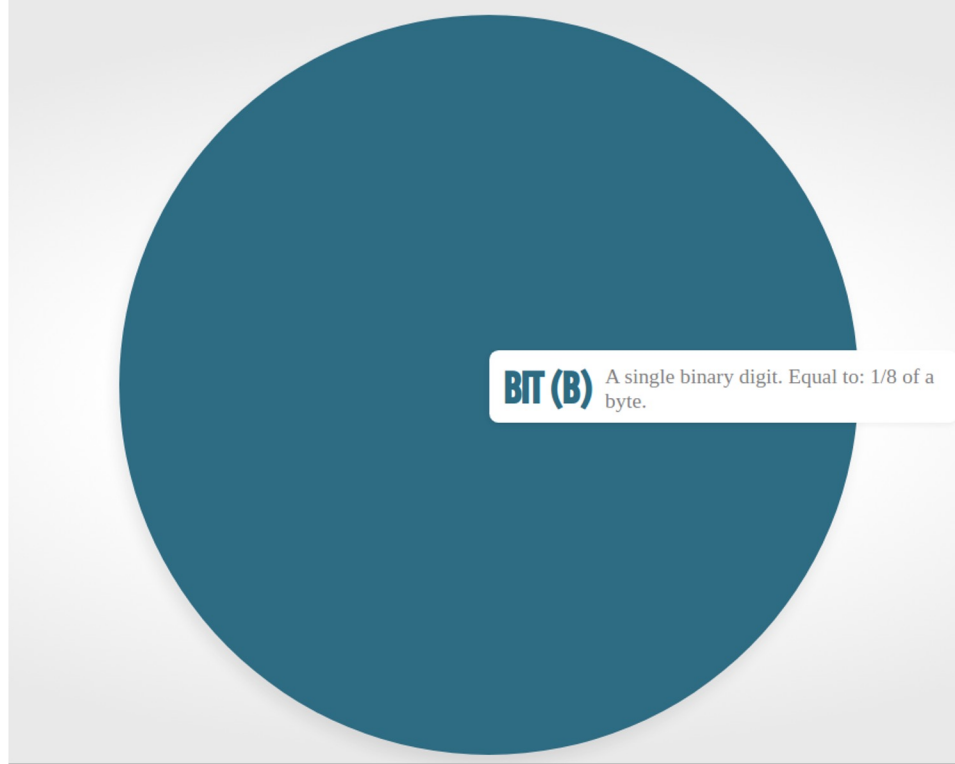
Binary number system has **base 2** and uses the digits 0 and 1.

Computers use the binary system and are able to use it to represent decimal numbers, text, images and audios.



Bit

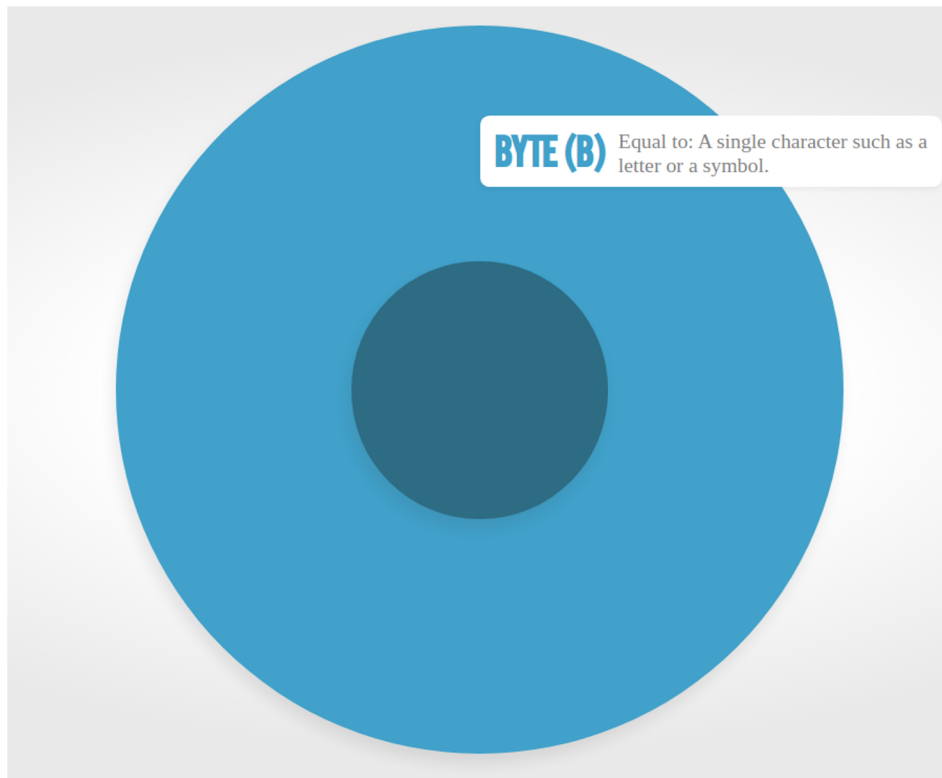
1
/
0



BIT (B) A single binary digit. Equal to: 1/8 of a byte.

Byte

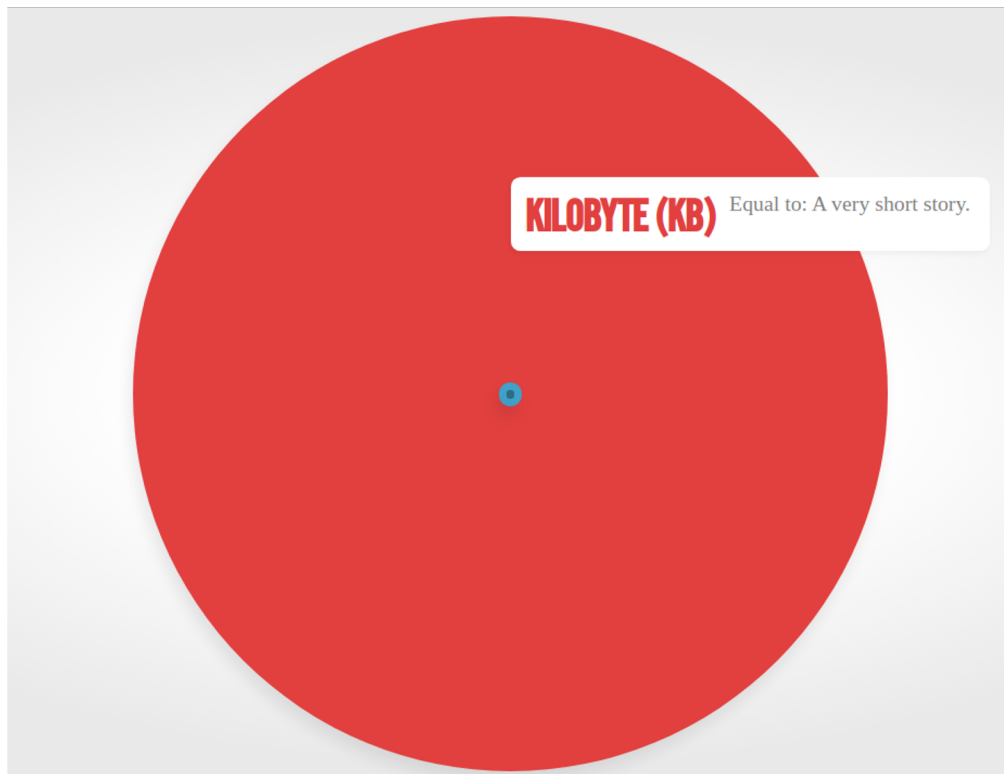
8 Bit



Redcentric

KiloByte

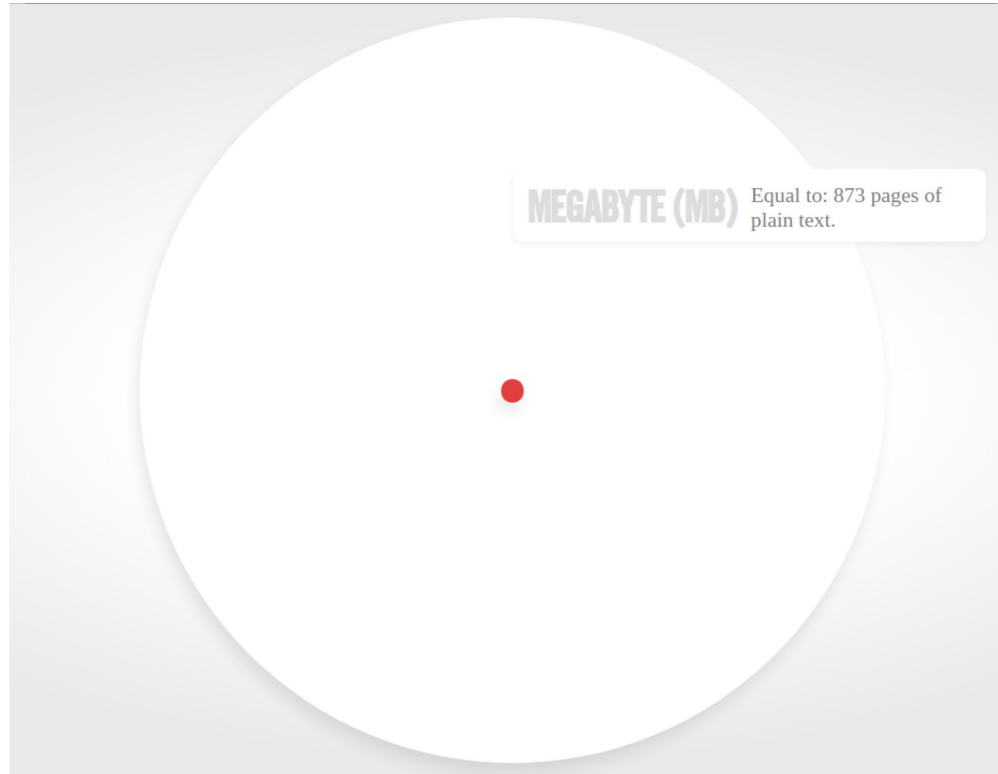
1,024 Bytes



Redcentric

MegaByte

1,048,576
Bytes



Redcentric

GigaByte

1,073,741,824
Bytes



From 0 to 16

Decimal	Binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111

1 bit
 2^1 values

2 bit
 2^2 values

3 bit
 2^3 values

Decimal	Binary
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

4 bit
 2^4 values



ASCII

With 1 byte we can represent $2^8 = 256$ values from 0 to 255.

ASCII codes used to represent text in computer are encoded with 1 byte.

Letter a	ASCII 097	Binary 01100001
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Letter b	ASCII 098	Binary 01100010
----------	-----------	-----------------

Letter A	ASCII 065	Binary 01000001
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RGB represent colors with the combination of 1 byte of Red 1 byte of Green 1 byte for Blue.



Decimal

A decimal number can be expressed as the multiplication of each digit with the base (10) to the power of the place occupied by the digit itself.

(DIGIT * BASE ^{POSITION})

1612 =

2 * 10⁰



Decimal

A decimal number can be expressed as the multiplication of each digit with the base (10) to the power of the place occupied by the digit itself.

(DIGIT * BASE^{POSITION})

1612 =

$$1 * 10^1 + 2 * 10^0$$



Decimal

A decimal number can be expressed as the multiplication of each digit with the base (10) to the power of the place occupied by the digit itself.

(DIGIT * BASE ^{POSITION})

$$1612 = 6 * 10^2 + 1 * 10^1 + 2 * 10^0$$



Decimal

A decimal number can be expressed as the multiplication of each digit with the base (10) to the power of the place occupied by the digit itself.

(DIGIT * BASE ^{POSITION})

$$1612 = 1 * 10^3 + 6 * 10^2 + 1 * 10^1 + 2 * 10^0$$



Decimal

A decimal number can be expressed as the multiplication of each digit with the base (10) to the power of the place occupied by the digit itself.

(DIGIT * BASE ^{POSITION})

$$1612 = 1 * 10^3 + 6 * 10^2 + 1 * 10^1 + 2 * 10^0$$

$$1612 = 1000 + 600 + 10 + 2$$



Binary to Decimal

To convert a binary number we can use the same process.

01100001 =

$1 * 2^0$



Binary to Decimal

To convert a binary number we can use the same process.

01100001 =

$$0 * 2^1 + 1 * 2^0$$



Binary to Decimal

To convert a binary number we can use the same process.

01100001 =

$$0 * 2^2 + 0 * 2^1 + 1 * 2^0$$



Binary to Decimal

To convert a binary number we can use the same process.

$$01100001 = 0 * 2^7 + 1 * 2^6 + 1 * 2^5 + 0 * 2^4 + 0 * 2^3 + 0 * 2^2 + 0 * 2^1 + 1 * 2^0$$

$$01100001 = 0 * 128 + 1 * 64 + 1 * 32 + 0 * 16 + 0 * 8 + 0 * 4 + 0 * 2 + 1 * 1$$



Binary to Decimal

To convert a binary number we can use the same process.

$$01100001 = 0 * 2^7 + 1 * 2^6 + 1 * 2^5 + 0 * 2^4 + 0 * 2^3 + 0 * 2^2 + 0 * 2^1 + 1 * 2^0$$

$$01100001 = 0*128 + 1*64 + 1*32 + 0*16 + 0*8 + 0*4 + 0*2 + 1*1$$

$$01100001 = 64 + 32 + 1$$

$$01100001 = 97$$



Exercise: Binary to Decimal

11010100



Exercise: Binary to Decimal

$$11010100 = 1 * 2^7 + 1 * 2^6 + 0 * 2^5 + 1 * 2^4 + 0 * 2^3 + 1 * 2^2 + 0 * 2^1 + 0 * 2^0$$

$$11010100 = 1*128 + 1*64 + 0*32 + 1*16 + 0*8 + 1*4 + 0*2 + 0*1$$

$$11010100 = 128 + 64 + 16 + 4 = \mathbf{212}$$



Decimal to Binary

To convert a decimal number into a binary we use the division method.

$$20 / 2 = 10 + 0$$

$$10 / 2 = 5 + 0$$

$$5 / 2 = 2 + 1$$

$$2 / 2 = 1 + 0$$

1) Divide the decimal number by 2, take the remainder and repeat these steps for the quotient.



Decimal to Binary

To convert a decimal number into a binary we use the division method.

$$20 / 2 = 10 + 0$$

$$10 / 2 = 5 + 0$$

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$$2 / 2 = 1 + 0$$

1) Divide the decimal number by 2, take the remainder and repeat these steps for the quotient.

2) At the end of the process starting from the last quotient, take all the remainders in reverse order to get the binary representation of the decimal

1 0 1 0 0

number

Exercise: Decimal to Binary

212



Exercise: Decimal to Binary

$$212 / 2 = 106 + 0$$

$$106 / 2 = 53 + 0$$

$$53 / 2 = 26 + 1$$

$$26 / 2 = 13 + 0 \quad 11010100$$

$$13 / 2 = 6 + 1$$

$$6 / 2 = 3 + 0$$

$$3 / 2 = 1 + 1$$



Exercise: Binary to Decimal

01000000



Exercise: Binary to Decimal

$$01000000 = 0 * 2^7 + 1 * 2^6 + 0 * 2^5 + 0 * 2^4 + 0 * 2^3 + 0 * 2^2 + 0 * 2^1 + 0 * 2^0$$

$$01000000 = 0 * 128 + 1 * 64 + 0 * 32 + 0 * 16 + 0 * 8 + 0 * 4 + 0 * 2 + 0 * 1$$

$$01000000 = \mathbf{64}$$



Exercise: Decimal to Binary

01000000



Exercise: Decimal to Binary

01000000 = **64**

$$\mathbf{64} / 2 = 32 + \mathbf{0}$$

$$32 / 2 = 16 + \mathbf{0}$$

$$16 / 2 = 8 + \mathbf{0}$$

$$8 / 2 = 4 + \mathbf{0}$$

$$4 / 2 = 2 + \mathbf{0}$$

$$2 / 2 = \mathbf{1} + \mathbf{0}$$

1000000 to store this value in one byte
01000000 add a zero in front.



Binary Summation

Rules:

$$0+0 = 0$$

$$1+0 = 1$$

$$0+1 = 1$$

$$1+1 = \mathbf{10} \text{ wait what?!}$$

1+1 is not 2. We need **carry** a bit to express the sum.

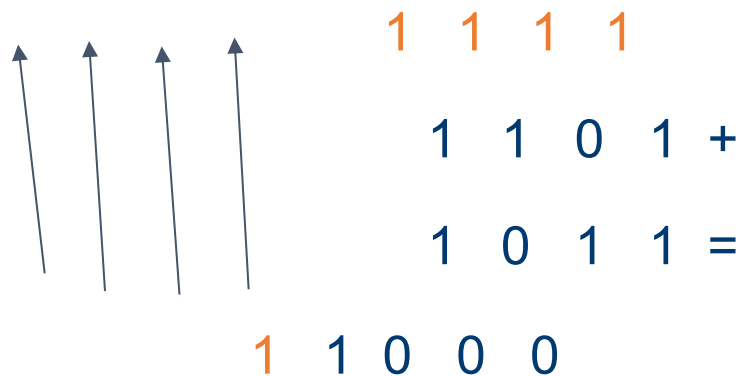
Bonus:

$$1 + 1 + 1 = 11$$



Binary Summation

1101 + 1011



The diagram illustrates the binary addition of 1101 and 1011. It shows the two numbers aligned vertically, with a plus sign to the right of the second number. Four arrows point upwards from the rightmost column to the left, indicating the propagation of a carry. The result of the addition is shown below the numbers, with an equals sign to the right. The carry values are highlighted in orange.

$$\begin{array}{r} 101 \\ +101 \\ \hline 1100 \end{array}$$

Binary Summation

(a) $11 + 1$

(b) $11 + 11$

(c) $111 + 11$

(d) $111 + 10$



Binary Summation

(a) $11 + 1 = 100$

(b) $11 + 11 = 110$

(c) $111 + 11 = 1010$

(d) $111 + 10 = 1001$



Binary Subtraction

Rules:

$$0-0 = 0$$

$$1-0 = 1$$

$$0-1 = ?$$

$$1-1 = 0$$

0-1 need to **borrow** a bit to express the subtraction.

Bonus:

$$10 - 1 = 1$$

$$11 - 1 = 10$$



Binary Subtraction

1101 - 1011



0 1

1 1 0 1 -

1 0 1 1 =

0 0 1 0



Binary Subtraction

(a) $11 - 10$

(b) $110 - 10$

(c) $1111 - 110$

(d) $100 - 10$



Binary Subtraction

(a) $11 - 10 = 1$

(b) $110 - 10 = 100$

(c) $1111 - 110 = 1001$

(d) $100 - 10 = 10$

